Problem A. A Good Problem

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	512 megabytes

A good problem should have a concise statement.

You are given an array a of length n, initially filled with zeros, and another array b of length n. Your goal is to transform array a into array b. You can perform the following two types of operations:

- 1 x: Add 1 to all elements in a that are equal to x.
- 2 x: Add 1 to the element in a at index x.

You can perform no more than 20 000 operations.

Input

The first line contains a positive integer n ($1 \le n \le 1000$).

The second line contains n non-negative integers representing the array b ($0 \le b_i \le n$).

Output

The first line should contain an integer k, representing the number of operations.

The following k lines should each contain two integers 1 x or 2 x, representing an operation. For the operation type 1 x, you must ensure that $0 \le x \le n$. For the operation type 2 x, you must ensure that $1 \le x \le n$.

Example

standard input	standard output
4 2 4 3 1	8 2 1 2 2 2 3 1 1 2 4 2 2 2 3 2 2